

Ilias Zadik, PhD

Department of Statistics and Data Science, Yale University
ilias.zadik@yale.edu

Current Appointment

- **Yale University** **New Haven, CT, USA**
Assistant Professor, Department of Statistics and Data Science 09/2023 – Present
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Past Academic Appointments

- **Massachusetts Institute of Technology (MIT)** **Cambridge, MA, USA**
Postdoctoral Associate, Department of Mathematics 09/2021 – 08/2023
Postdoctoral mentors: Elchanan Mossel and Nike Sun
 - **New York University (NYU)** **New York, NY, USA**
CDS Moore-Sloan Postdoctoral Fellow (Faculty Fellow), Center for Data Science (CDS) 09/2019 – 08/2021
Conducted independent research.
Co-instructed two graduate level classes in Data Science.
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Research Interests

- High dimensional statistics, foundations of data science/machine learning, probability theory, algorithms, differential privacy.
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Education

- **Massachusetts Institute of Technology (MIT)** **Cambridge, MA, USA**
PhD, Operations Research. GPA: 5.0/5.0 09/2014 – 09/2019
PhD advisor: David Gamarnik.
Thesis: “Computational and Statistical Challenges in High Dimensional Statistical Models”
 - **Trinity College, Cambridge University** **Cambridge, UK**
M.A.St. in Mathematics, Part III, with Distinction (ranked 13th out of 247 students) 09/2013 – 07/2014
Part III Essay: “Noise Sensitivity with applications to Percolation and Social Choice Theory”
Part III Essay advisor: Béla Bollobás
 - **University of Athens** **Athens, Greece**
B.A. in Mathematics, Graduated with GPA 10.0/10.0 09/2009 – 02/2013
First known *perfect* GPA achieved in the recorded history of the department.
Undergraduate research advisor: Vassili Nestoridis. Research area: Complex Analysis.
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Industry Experience

- **Microsoft Research New England** **Cambridge, MA, USA**
Summer Internship 06/2017 – 08/2017
Mentored by Jennifer Chayes and Christian Borgs.
Performed research on differential privacy and large networks (led to conference paper (C26)).
Also collaborated with Lester Mackey and Vasilis Syrgkanis (led to conference paper (C27)).
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Research

Note: As customary in my field, the order of the authors is **alphabetical**.
Few exceptions are denoted by (*).

Journal Papers (Accepted or Publications)

- (J1) "Almost-Optimal Local-Search Methods for Sparse Tensor PCA";
Max Lovig, Conor Sheehan, Konstantinos Tsirkas, Ilias Zadik
Annals of Applied Probability (accepted, 2026).
- (J2) "On The MCMC Performance In Bernoulli Group Testing And The Random Max Set Cover Problem";
Max Lovig, Ilias Zadik
Annals of Applied Probability (accepted, 2026).
- (J3) "Counting Stars is Constant-Degree Optimal For Detecting Any Planted Subgraph";
Xifan Yu, Ilias Zadik, Peiyuan Zhang.
Mathematical Statistics and Learning (accepted, 2025)
Preprint available at <https://arxiv.org/abs/2403.17766>.
Journal version of conference paper (C7).
- (J4) "A Bayesian Proof of the Spread Lemma";
Elchanan Mossel, Jonathan Niles-Weed, Nike Sun, Ilias Zadik
Random Structures and Algorithms (accepted, 2025)
Preprint available at <https://arxiv.org/abs/2209.11347>.
- (J5) "Sharp Thresholds Imply Circuit Lower Bounds";
David Gamarnik, Elchanan Mossel, Ilias Zadik
Israel Journal of Mathematics (accepted, 2025)
Preprint available at <https://arxiv.org/abs/2311.04204>.
- (J6) "Almost-Linear Planted Cliques Elude the Metropolis Process";
Zongchen Chen, Elchanan Mossel, Ilias Zadik
Random Structures and Algorithms, 2025, Volume 66, pp 1-24
Preprint available at <https://arxiv.org/abs/2204.01911>.
Journal version of conference paper (C9).
- (J7) "Sharp thresholds in inference of planted subgraphs";
Elchanan Mossel, Jonathan Niles-Weed, Youngtak Sohn, Nike Sun, Ilias Zadik.
Annals of Applied Probability, 2025, Volume 35, pp 523-563
Preprint available at <https://arxiv.org/abs/2302.14830>.
Journal version of conference paper (C8).
- (J8) "Stationary Points of Shallow Neural Networks with Quadratic Activation Function";
David Gamarnik, Eren Kizildag, Ilias Zadik
Mathematics of Operations Research, 2025, Volume 50, pp 1-781
Preprint available at <https://arxiv.org/abs/1912.01599>.
- (J9) "The Landscape of the Planted Clique Problem: Dense Subgraphs and the Overlap Gap Property";
David Gamarnik, Ilias Zadik.
Annals of Applied Probability, 2024, Volume 34, pp 3375-3434
Preprint available at <https://arxiv.org/abs/1904.07174>.

- (J10) "Shapes and recession cones in mixed-integer convex representability"
 Ilias Zadik, Miles Lubin, Juan Pablo Vielma (*)
Mathematical Programming, 2024, Volume 204, pp 739–752
 Preprint available at <https://arxiv.org/abs/2103.03379>.
- (J11) "Free Energy Wells and the Overlap Gap Property in Sparse PCA" ;
 Gérard Ben Arous, Alexander Wein, Ilias Zadik.
Communications on Pure and Applied Mathematics, 2023, Volume 76, pp 2410-2473
 Preprint available at <https://arxiv.org/abs/2006.10689>.
 Journal version of conference paper (C18).
- (J12) "It was "all" for "nothing": sharp phase transition for noiseless discrete channels. ";
 Jonathan Niles-Weed, Ilias Zadik.
IEEE Transactions of Information Theory, 2023, Volume 69, pp 5188-5202
 Preprint available at <https://arxiv.org/abs/2102.12422>.
 Journal version of conference paper (C13).
- (J13) "Sparse High-Dimensional Regression. Algorithmic Barriers and a Local Search Algorithm";
 David Gamarnik, Ilias Zadik.
The Annals of Statistics, 2022, Vol. 50, pp 880-903
 Journal version of conference paper (C26).
- (J14) "Mixed integer convex representability";
 Miles Lubin, Juan Pablo Vielma and Ilias Zadik
Mathematics of Operations Research, 2021 Vol. 47, pp 720-749.
 Journal version of conference paper (C27).
- (J15) "The All-or-Nothing Phenomenon in Sparse Linear Regression";
 Galen Reeves, Jiaming Xu, Ilias Zadik.
Mathematical Statistics and Learning, 2021, Vol. 3, pp 259-313
 Journal version of conference paper (C20).
- (J16) "Inference in High-Dimensional Linear Regression via Lattice Basis Reduction and Integer Relation Detection";
 David Gamarnik, Eren C. Kizildag, Ilias Zadik
IEEE Transactions of Information Theory, 2019, Vol 67, pp 8109 - 8139.
 Journal version of conference paper (C23).
- (J17) "Self-Regularity of Non-Negative Output Weights for Overparameterized Two-Layer Neural Networks"
 David Gamarnik, Eren C. Kızıldağ, Ilias Zadik
IEEE Transactions on Signal Processing, 2022, Vol. 70, pp 1310–1319.
 Journal version of conference paper (C15).
- (J18) "Universal Padé Approximants and their behaviour on the boundary"
 Ilias Zadik
Monatshefte für Mathematik, 2017, Vol. 182, pp 173–193.
- (J19) "Padé approximants, density of rational functions in $A^\infty(V)$ and smoothness of the integration operator";
 Vassili Nestoridis, Ilias Zadik
Journal of Mathematical Analysis and Applications, 2015, Vol. 423, pp 1514-1539.

Peer-reviewed Conference Papers

- (C1) "The monotonicity of the Franz-Parisi potential is equivalent with Low-degree MMSE lower bounds";
Konstantinos Tsirkas, Leda Wang, Ilias Zadik
In Proceedings of the *Conference on Learning Theory* (COLT), 2026.
- (C2) "Stable Algorithms Lower Bounds For Estimation From MMSE Discontinuities";
Xifan Yu, Ilias Zadik
In Proceedings of the *Conference on Learning Theory* (COLT), 2026.
- (C3) "An Optimized Franz-Parisi Criterion and its Equivalence with SQ Lower Bounds";
Siyu Chen, Theodor Misiakiewicz, Ilias Zadik, Peiyuan Zhang
In Advances of the 38th *Neural Information Processing Systems* (NeurIPS), 2025 (**Oral presentation**).
- (C4) "The Fundamental Limits of Recovering Planted Subgraphs";
Daniel Lee, Francisco Pernice, Amit Rajaraman, Ilias Zadik.
In Proceedings of the *Conference on Learning Theory* (COLT), 2025.
- (C5) "On the Hardness of Learning One Hidden Layer Neural Networks";
Shuchen Li, Ilias Zadik, Manolis Zampetakis.
In Proceedings of the Conference on *Algorithmic Learning Theory* (ALT), 2025.
- (C6) "Low-degree Security of the Planted Random Subgraph Problem";
Andrej Bogdanov, Chris Jones, Alon Rosen, Ilias Zadik.
In Proceedings of the *Theory of Cryptography Conference* (TCC), 2024.
- (C7) "Counting Stars is Constant-Degree Optimal For Detecting Any Planted Subgraph";
Xifan Yu, Ilias Zadik, Peiyuan Zhang.
In Proceedings of the *Conference on Learning Theory* (COLT), 2024.
- (C8) "Sharp thresholds in inference of planted subgraphs";
Elchanan Mossel, Jonathan Niles-Weed, Youngtak Sohn, Nike Sun, Ilias Zadik.
In Proceedings of the *Conference on Learning Theory* (COLT), 2023.
- (C9) "Almost-Linear Planted Cliques Elude the Metropolis Process";
Zongchen Chen, Elchanan Mossel, Ilias Zadik
In Proceedings of the *Symposium on Discrete Algorithms* (SODA), 2023.
- (C10) "Archimedes Meets Privacy: On Privately Estimating Quantiles in High Dimensions Under Minimal Assumptions";
Omri Ben-Eliezer, Dan Mikulincer, Ilias Zadik
In Advances of the 35th *Neural Information Processing Systems* (NeurIPS), 2022.
- (C11) "The Franz-Parisi Criterion and Computational Trade-offs in High Dimensional Statistics";
Afonso Bandeira, Ahmed El Alaoui, Sam Hopkins, Tselil Schramm, Alexander S Wein, Ilias Zadik
In Advances of the 35th *Neural Information Processing Systems* (NeurIPS), 2022 (**Oral presentation**).
- (C12) "Lattice-based methods surpass sum-of-squares in clustering" ;
Ilias Zadik, Min Jae Song, Alexander S. Wein, Joan Bruna (*)
In Proceedings of the 35th *Conference in Learning Theory* (COLT), 2022, pp1247-1248.
- (C13) "Statistical and Computational Phase Transitions in Group Testing";
Amin Coja-Oghlan, Oliver Gebhard, Max Hahn-Klimroth, Alexander S Wein, Ilias Zadik
In Proceedings of the 35th *Conference in Learning Theory* (COLT), 2022, pp 4764-4781.

- (C14) "On the cryptographic hardness of learning single periodic neurons";
Min Jae Song, Ilias Zadik, Joan Bruna (*)
In *Advances of the 34th Neural Information Processing Systems (NeurIPS)*, 2021, pp 29602-29615
- (C15) "It was 'all' for 'nothing': sharp phase transition for noiseless discrete channels. ";
Jonathan Niles-Weed, Ilias Zadik.
In *Proceedings of the 34th Conference in Learning Theory (COLT)*, 2021, pp 3546-3547.
- (C16) "Group testing and local search: is there a computational-statistical gap?";
Fotis Iliopoulos, Ilias Zadik.
In *Proceedings of the 34th Conference in Learning Theory (COLT)*, 2021 pages 2499-2551.
- (C17) "Self-Regularity of Non-Negative Output Weights for Overparameterized Two-Layer Neural Networks";
David Gamarnik, Eren Kizildag, Ilias Zadik
In *Proceedings of the IEEE International Symposium of Information Theory (ISIT)*, 2021, pages 819-824.
- (C18) "The All-or-Nothing Phenomenon in Sparse Tensor PCA ";
Jonathan Niles-Weed, Ilias Zadik.
In *Advances of the 34th Neural Information Processing Systems (NeurIPS)* 2020
- (C19) "Optimal Private Median Estimation under Minimal Distributional Assumptions ";
Christos Tzamos, Emmanouil Vlatakis, Ilias Zadik.
In *Advances of the 34th Neural Information Processing Systems (NeurIPS)* 2020 **(Spotlight)**
- (C20) "Free Energy Wells and the Overlap Gap Property in Sparse PCA" ;
Gérard Ben Arous, Alexander Wein, Ilias Zadik.
In *Proceedings of the 33rd Conference on Learning Theory (COLT)* 2020, pages 479-482.
- (C21) "All-or-Nothing Phenomena: From Single-Letter to High Dimensions";
Galen Reeves, Jiaming Xu, Ilias Zadik.
In *Proceedings of the 8th IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing, (CAMSAP)* 2019, pages 654-658.
- (C22) "The All-or-Nothing Phenomenon in Sparse Linear Regression";
Galen Reeves, Jiaming Xu, Ilias Zadik
In *Proceedings of the 32nd Conference on Learning Theory (COLT)* 2019, pages 2652-2663.
- (C23) "Improved bounds on Gaussian MAC and sparse regression via Gaussian inequalities";
Ilias Zadik, Christos Thrampoulidis, Yury Polyanskiy (*)
In *Proceedings of the IEEE International Symposium of Information Theory (ISIT)* 2019, pages 430-434.
- (C24) "A Simple Bound On The BER Of The MAP Decoder For Massive MIMO System";
Christos Thrampoulidis, Ilias Zadik, Yury Polyanskiy (*)
In *Proceedings of the 44th International Conference on Acoustics, Speech, and Signal Processing (ICASSP)* 2019, pages 4544-4548.
- (C25) "High-Dimensional Linear Regression using Lattice Basis Reduction"
David Gamarnik, Ilias Zadik
In *Advances of the 32nd Neural Information Processing Systems (NeurIPS)* 2018, pages 1842–1852.
- (C26) "Revealing Network Structure Confidentially: Improved Rates for Node-Private Graphon Estimation";
Christian Borgs, Jennifer Chayes, Adam Smith, Ilias Zadik.
In *Proceedings of the 59th IEEE Annual Symposium on Foundations of Computer Science (FOCS)* 2018, pages 533–543.

- (C27) "Orthogonal Machine Learning: Power and Limitations";
Lester Mackey, Vasilis Sygrkanis, Ilias Zadik.
In Proceedings of the 35th *International Conference of Machine Learning* (ICML) 2018, pages 5723–5731.
- (C28) "High-dimensional Regression with Binary Coefficients. Estimating Squared error and the Phase Transition Property" ;
David Gamarnik, Ilias Zadik.
In Proceedings of the 30th *Conference on Learning Theory* (COLT) 2017, pages 948–953.
- (C29) "Mixed integer convex representability";
Miles Lubin, Ilias Zadik, Juan Pablo Vielma (*)
In Proceedings of the 19th *Integer Programming and Combinatorial Optimization* conference (IPCO) 2017, pages 392–404.

Preprints/In Submission Papers

- (P1) "A Bayesian proof of the Bernoulli theorem";
Jingbo Liu, Ilias Zadik
Preprint available at <https://arxiv.org/abs/2608.11031>.
- (P2) "A Bayesian proof and interpretation of Talagrand's Majorizing measure theorem";
Ilias Zadik
Preprint available at <https://arxiv.org/pdf/2605.30321>.
- (P3) "Node-Private Community Detection in Stochastic Block Models";
Olga Klopp, Ilias Zadik
Preprint available at <https://arxiv.org/pdf/2604.09078>.
- (P4) "Near-optimal node-private community estimation in polynomial-time";
Laurentiu Marchis, Olga Klopp, Po-Ling Loh, Ilias Zadik (*)
Preprint available at <https://arxiv.org/pdf/2607.09441>.
- (P5) "On the Low-Temperature MCMC threshold: the cases of sparse tensor PCA, sparse regression, and a geometric rule";
Zongchen Chen, Conor Sheehan, Ilias Zadik
Preprint available at <https://arxiv.org/abs/2408.00746>.
- (P6) "Transfer Learning Beyond Bounded Density Ratios";
Alkis Kalavasis, Ilias Zadik, Manolis Zampetakis.
Preprint available at <https://arxiv.org/abs/2403.11963>.

Books and Surveys

- (B1) "An Introduction to Computational-Statistical Tradeoffs in High-Dimensional Statistics";
Ilias Zadik.
Invited for submission in *Foundations and Trends in Statistics*, July 2026.
June 2026 draft available at my personal website.
- (B2) "Noise Sensitivity with applications to Percolation and Social Choice Theory";
Ilias Zadik (advised by Bela Bollobas)
Part III (Master) Essay, May 2014
Preprint available at personal website
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Invited Talks

- (T57) International Conference on Non-parametric Statistics (ISNPS), Thessaloniki, June 2026
- (T56) Stochastic Networks, Applied Probability, and Performance (SNAPP) Seminar, April 2026
- (T55) Harvard University, Statistics Department, Probability Seminar, October 2025
- (T54) INFORMS Annual meeting, Georgia Tech, October 2025
- (T53) Workshop "Statistical Physics & Machine Learning: moving forward", Cargese, August 2025
- (T52) INFORMS Applied Probability meeting, Georgia Tech, June 2025
- (T51) Workshop "Computational-Statistical Trade-offs", TTIC, June 2025
- (T50) University of Wisconsin, Madison, Computer Science Department, March 2025
- (T49) Workshop "Towards a theory for typical case algorithmic hardness",
Les Houches Physics School, France, February 2025
- (T48) Columbia University, Statistics Department, January 2025
- (T47) Caltech, AIM workshop on "Low-degree polynomials", December 2024
- (T46) International Symposium on Mathematical Programming, bi-Annual meeting, July 2024
- (T45) Bocconi University, Cryptography and Average-case analysis workshop, May 2024
- (T44) Georgia Tech, "Probability, Algorithms and Inference" summer school, May 2024
- (T43) AMS meeting on "Computational-Statistics Trade-offs", University of San Francisco, March 2024
- (T42) New York University (Courant), Discrete Mathematics Seminar, February 2024
- (T41) Lehigh University and University of Minnesota, joint probability seminar, November 2023
- (T40) University of Athens, Department of Mathematics seminar, November 2023
- (T39) University of Texas, Austin, Institute for Foundations of Machine Learning seminar, November 2023
- (T38) Johns Hopkins University, Applied Mathematics and Statistics seminar, October 2023
- (T37) Workshop "Statistical Physics & Machine Learning back together again", Cargese, August 2023
- (T36) Workshop "Connecting Physics, Geometry, and Algebraic Hardness", Santa Fe Institute, July 2023
- (T35) Workshop on "Graphical Models, Statistical Inference and Applications", Harvard, May 2023
- (T34) AMS meeting on "High-dimensional Convexity and Probability", Georgia Tech, March 2023
- (T33) BIRS workshop on "Learning in Networks: Performance Limits and Algorithms", November 2022
- (T32) Cornell University, Statistics Seminar, November 2022
- (T31) Worcester Polytechnic Institute, Computer Science Colloquium, November 2022
- (T30) UC Davis, Probability Seminar, October 2022
- (T29) INFORMS Annual Meeting, October 2022

- (T28) New York University, Mathematics, Information and Computation (MIC) Seminar, October 2022
 - (T27) Simons workshop on “Graph Limits, Nonparametric Models, and Estimation”, September 2022
 - (T26) MIT, IDSS Statistics Seminar, February 2022
 - (T25) Simons workshop on “Cryptography and Learning”, November 2021
 - (T24) Joint Simons/IFML workshop, Berkeley, October 2021
 - (T23) INFORMS Annual Meeting, October 2021
 - (T22) Simons workshop on “Rigorous Evidence for Computational-Statistical Trade-offs”, September 2021
 - (T21) BIRS workshop on "Random Graphs and Statistical Inference", August 2021
 - (T20) New York Colloquium on Algorithms and Complexity (NYCAC), March 2021
 - (T19) 19th Northeast Probability Seminar, November 2020
 - (T18) INFORMS Annual Meeting, November 2020
 - (T17) Simons workshop on “Computational Phase Transitions”, September 2020
 - (T16) IBM Thomas J Watson Research Center, February 2020
 - (T15) Google Research Algorithm’s Seminar, NYC, February 2019
 - (T14) New York University, Mathematics, Information and Computation (MIC) Seminar, February 2019
 - (T13) Stanford, Theory of Computer Science Seminar, January 2019
 - (T12) Northeastern University, Theory of Computer Science Seminar, November 2018
 - (T11) Microsoft Research New England, Machine Learning (ML) Ideas Lunch, November 2018
 - (T10) Cornell University, workshop for young Operations Research researchers, October 2018
 - (T9) MIT LIDS and Statistics Tea talk, April 2018
 - (T8) Oberwolfach’s workshop on "Network Models: Structure and Functions", December 2017
 - (T7) INFORMS Annual Meeting, November 2017
 - (T6) INFORMS Applied Probability Society Conference, July 2017
 - (T5) Integer Programming and Combinatorial Optimization Conference, July 2017
 - (T4) MIT LIDS and Statistics Tea talk, March 2017
 - (T3) MIT LIDS student seminar, January 2017
 - (T2) University of Athens Probability and Statistics Seminar, January 2017
 - (T1) MIT Operations Research Center Student Seminar, November 2016
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Current/past PhD students mentoring at Yale University

- Max Lovig, 4th year PhD student at Yale University, Statistics and Data Science department.
Co-advised with Professor Zhou Fan.
 - Conor Sheehan, 4th year PhD student at Yale University, Statistics and Data Science department.
 - Kostas Tsirkas, 2nd year PhD student at Yale University, Statistics and Data Science department.
 - Shuchen Li, 2nd year PhD student at Yale University, Computer Science department.
Co-advised with Professor Manolis Zampetakis.
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Current/past visiting students in my research group (including Fall 2026)

- Aleix Artigas More (March 2026 - September 2026)
6-months research visiting undergraduate student from CFIS-Universitat Politècnica de Catalunya.
Ongoing research project in the area of computational-statistical trade-offs.
 - Orestis Lignos (October 2026 - December 2026)
2-months research visiting undergraduate student from University of Athens, Greece.
Ongoing research project in the area of random graphs.
 - Paula Murmann (September 2026 - February 2027)
6-months research visiting PhD student from EPFL.
Ongoing research project in the area of Bayesian inference and epidemics models.
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PhD dissertation committees (beyond my PhD students)

- Xifan Yu, Yale, Computer Science (expected, Spring 2027).
Advisor: Professor Spielman.
 - Timothy Wee, Yale, Statistics and Data Science (Spring 2024).
Advisor: Professor Tatikonda.
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Reading Groups at Yale

- Research Group meeting (September 2026 - Present).
One hour weekly meetings with the members of my research group.
Each time discussing a new state-of-the-art result in our research area.
 - Reading Group on Machine Learning theory (Fall 2024 & Fall 2025)
Co-organized with Yale Professors Chewi, Misiakiewicz, Montasser.
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Teaching Experience (Yale)

- Spring 2024 & Spring 2025 & Spring 2026: S&DS 351 (Yale), "Stochastic Processes"
Instructor.
Undergraduate and Graduate-level class on topics such as:
Markov chains, martingales, Brownian motion

- Fall 2023 & Fall 2024: S&DS 688 01 (Yale), "Computational-Statistical Trade-offs."
Instructor.
Newly designed research-level class on recent research topics such as:
replica-symmetric predictions, low-degree lower bounds, SQ lower bounds, MCMC lower bounds.
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Teaching Experience (before Yale)

- Fall 2020: DS-GA 1005 (NYU), "Inference and Representation."
Co-instructor with Joan Bruna. We redesigned together the class this year.
Graduate-level advanced class on topics such as:
graphical models, variational inference, MCMC sampling and optimal transport.
 - Fall 2019, DS-GA 1002 (NYU), "Probability and Statistics for Data Science."
Co-instructor with Carlos Fernandez-Granda.
Graduate-level introductory class on probability and statistics.
 - Spring 2017, 15.070J/6.265J (MIT), "Modern Discrete Probability."
Teaching assistant
Duties included: weekly office hours, assignment preparation and grading.
Graduate-level class taught by Yury Polyanskiy and Guy Bresler.
 - Fall 2016, 15.085J/6.436J (MIT), "Fundamentals of Probability."
Teaching assistant.
Duties included: weekly office hours, recitations, assignment preparation and grading.
Graduate-level class taught by David Gamarnik.
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Committees at Yale University

- Fall 2024 & 2025 & 2026: PhD admission committee for the Statistics and Data Science department.
 - Fall 2025: Tenure-track search committee for the Statistics and Data Science department.
 - Fall 2025: Reappointment committee for Yale faculty, Professor Jay Emerson.
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Service and Outreach (Yale and beyond)

- Co-organizing the three-day Yale Foundations of Data Science workshop "Probability, Computation, and Learning in High Dimensions" at Yale University, May 2027
- Currently serving in the 2026 INFORMS APS Best Student Paper Prize committee.
- COLT 2026 2-hours research tutorial on "Equivalences between lower bounds frameworks: Low-degree polynomials and physics predictions", San Diego, June 2026.
- Georgia Tech Summer School 2-hours research tutorial on "Sharp thresholds in inference and implications on combinatorics and circuit lower bounds", Atlanta, May 2024.
- Served in the *program committee* for the Conference on Learning theory (COLT) 2021, 2022, 2023, 2024, 2025, 2026 and in the *senior program committee* for COLT 2024, 2025, 2026.
COLT is the leading conference for publications in learning theory.
- Served in the *program committee* for International Colloquium on Automata, Languages and Programming (ICALP) 2026.
ICALP is among the leading conferences for publications in theoretical computer science.

- Co-organizing the Foundations of Data Science Seminar Series, Yale University for Fall 2026.
- Co-organized the Department of Statistics and Data Science Seminar Series, Yale University for both years 2023 and 2024.
- Co-organized the (*virtual and open to the public*) “*Math and Data (MaD) +*” seminar.
The MaD+ seminar took place from Spring 2020 to Spring 2021.
Our goal was to assist research on the foundations of data science during the COVID-19 pandemic.
For more information, please see the seminar’s website.
- Served (multiple times) as a reviewer for many journals including for:
 - Annals of Statistics
 - Communications on Pure and Applied Mathematics
 - Mathematical Statistics and Learning
 - Probability Theory and Related Fields
 - Annals of Applied Probability
 - Mathematical Programming
 - SIAM Journal of Discrete Mathematics
 - SIAM Journal on Mathematics of Data Science
 - SIAM Journal of Optimization
 - Combinatorica
 - Operations Research
 - Journal of Machine Learning Research
 - IEEE Journal on Selected Areas in Information Theory.
- Served (multiple times) as a reviewer/sub-reviewer for many conferences including for:
 - COLT,
 - NeurIPS
 - FOCS
 - STOC
 - ITCS
 - ISIT
 - ICALP
 - SODA

Awards and Honors

- My paper (J8) was selected as *the 2025 highlight paper*, for the 50 papers-50 years list prepared by the senior editors of the Mathematics of Operations Research journal.
- *Oral* NeurIPS presentations for my two papers (C3), (C11) on the Franz-Parisi potential.
- CDS Moore-Sloan *Postdoctoral Fellowship* at NYU, 2019-2021.
- Top 400 Reviewers Award, for reviewing for Neural Information Processing Systems (NeurIPS), 2019.

- *Honorable Mention* for MIT Operations Research Center Best Student Paper Award, 2017
Awarded for paper (J13), now published in the Annals of Statistics, 2022.
- *Senior scholarship* from Trinity College, Cambridge University, 2014.
Awarded for achieving “Distinction” in Part III examination.
- The Onassis Foundation Scholarship for my Master’s studies, 2013-2014.
- The Cambridge Home and European Scholarship Scheme (CHESS) award for my Master’s studies, 2013-2014.
- IKY scholarship for top academic performance in *each year* of my undergraduate studies, 2009-2012.
- International Mathematics Competition for university students (IMC):
First Prize, 2011; Second Prize, 2010.
- South Eastern European Mathematical Olympiad for University students (SEEMOUS):
Gold Medal (scored 39.5/40 and ranked 1st), 2011 ; Silver Medal, 2010.
- International Mathematical Olympiad (IMO): *Honorable Mention, 2009.*